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CHAPTER 1 • NUMBER • TOPIC 7

Compound Interest & Depreciation

Each year the percentage acts on the new amount — so one multiplier, raised to the number of years: $\text{value} = P \times m^n$.



$$\frac{3}{10} = 0.3 =$$

30%**5**

VARIANTS

29

PAST-PAPER PARTS

24

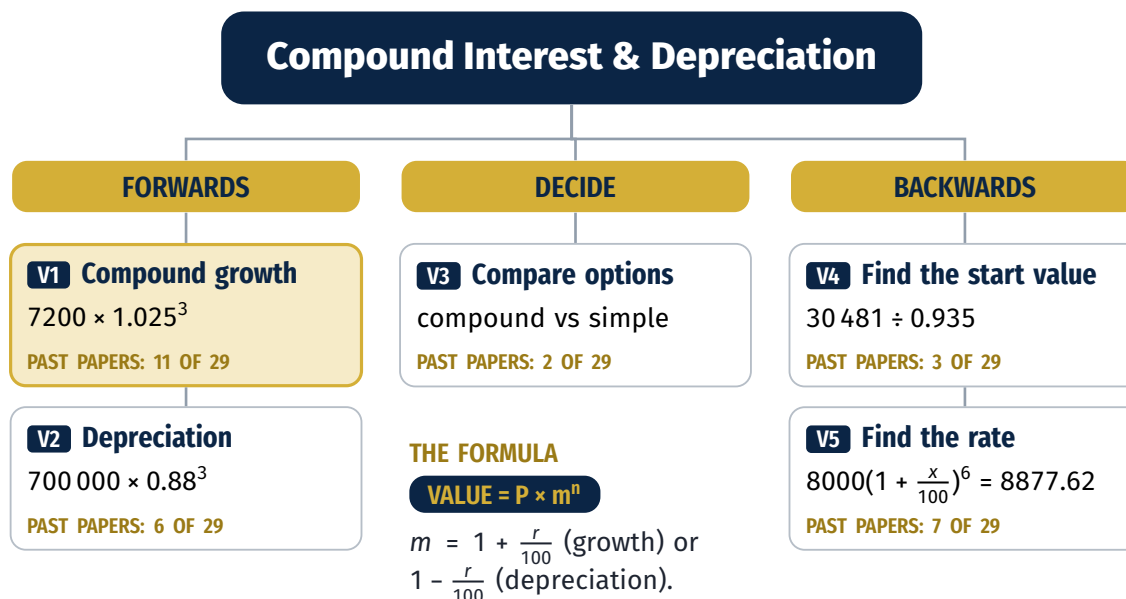
PRACTICE QUESTIONS

YOU WILL MASTER**V1** Compound growth**V2** Depreciation**V3** Compare options**V4** Find the start value**V5** Find the rate

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Compound Interest at a glance



Your notes


★ How often does each variant come up?

26 past-paper questions (Jan 2020 → Nov 2025) hold 29 parts; 24 practice questions — grouped by type.

	PAST PAPERS (OUT OF 29)	PRACTICE PARTS
V1 Compound growth	11	6
V2 Depreciation	6	1
V3 Compare options	2	6
V4 Find the start value	3	3
V5 Find the rate	7	8

WHAT THIS TELLS YOU A compound-interest question is on nearly every Paper 2: 17 of 29 parts are “value after n years” (V1, V2) — 3 calculator marks. The harder ones (V5) ask for the rate.

BEFORE YOU START

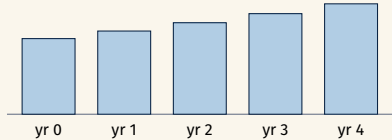
The big idea



Interest on the interest.

Simple interest: the same amount every year (on the start only).

Compound interest: each year's percentage is of the **new** amount — so it grows faster. One multiplier, used once per year.



×1.1 each year: the bars grow by more each time

The formula

$$\text{value} = P \times m^n$$

P = starting amount m = multiplier (1.025 for +2.5%, 0.88 for -12%)
 n = number of years

Growth

$+r\% \rightarrow m = 1 + \frac{r}{100}$
 interest, appreciation

Depreciation

$-r\% \rightarrow m = 1 - \frac{r}{100}$
 value falls

Interest only

value - P
 "how much interest?"

**CALCULATOR FIRST, THEN CHECK**

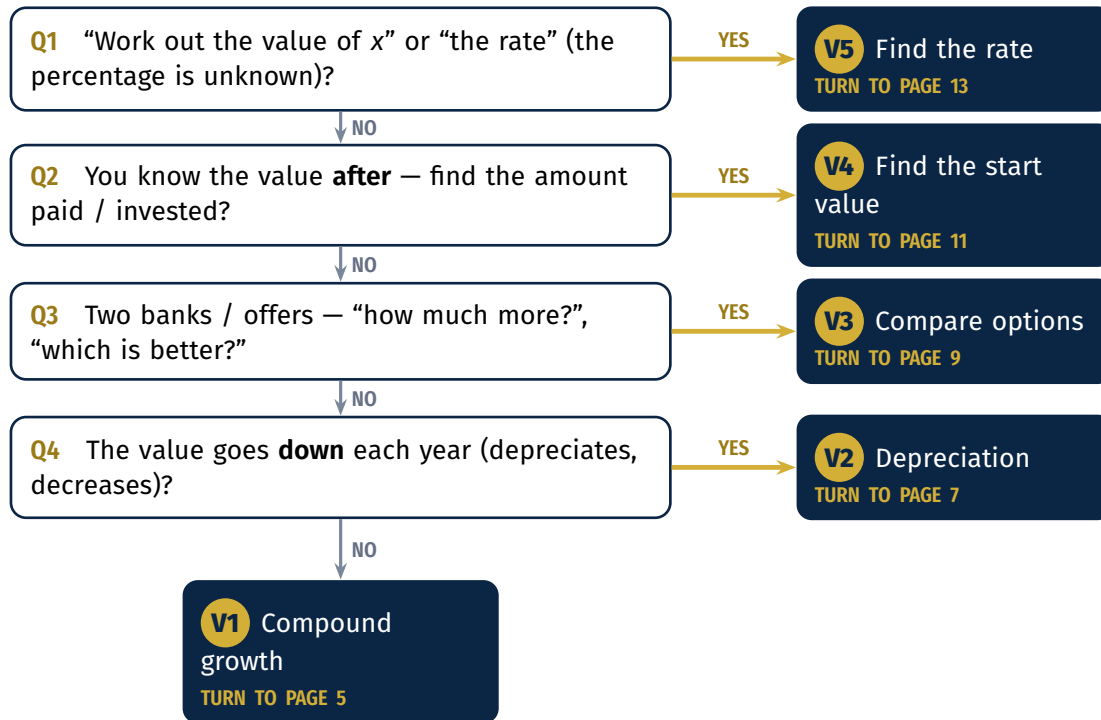
One calculator line: $7200 \times 1.025 \times^{\square} 3 =$. For a rate, use **SHIFT** **CALC** (SOLVE) on the equation, or take a root: $\sqrt[3]{8877.62 \div 8000}$. Round only at the end.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?

 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant – and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------|
|  | What is it? |  | When you see → you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1

VARIANT 1 MOST COMMON IN THE EXAM

Compound Growth

$$\text{Value} = P \times \left(1 + \frac{r}{100}\right)^n$$

SEEN IN PAST PAPERS

11 / 29

PRACTICE QUESTIONS: 6



Your notes

Came up in: Oct/Nov 2020 P1HR Q8 • Jan 2021 P1HR Q7 • Jan 2021 P1H Q7(a) • May 2021 P2H Q11(b) • Jan 2022 P2HR Q5 • Jun 2022 P2H Q9 • Jun 2023 P2H Q8 • May 2024 P1HR Q5 • Jun 2024 P2H Q9 • Nov 2024 P2H Q9 • Jun 2025 P2H Q7

? What is it?

In plain words

Each year multiply by the **same multiplier**. 3 years → multiply three times → **power 3**.

7200 at 2.5% for 3 years:

$$7200 \times 1.025^3$$



$$= 7200 \times 1.025^3$$

IT LOOKS LIKE

savings account

painting increases by 4%

How do I do it?

1 Multiplier

+2.5% → 1.025.

2 Power

Number of years → the power: 1.025^3 .

3 Multiply

$P \times m^n$ in one calculator line.

4 Round

To the unit asked (nearest dollar, yuan ...).

$P \times m^n$ Not $P \times m \times n$.

V1 Compound Growth

VARIANT 1 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“ $x\%$ per year compound interest ... end of 3 years” → $P \times \left(1 + \frac{x}{100}\right)^3$

“The value increases by 4% each year” → **Same method:** $\times 1.04^n$

“Give your answer to the nearest dollar” → **Round only the final value**

WORKED EXAMPLE **PAST PAPER**

January 2022 • Paper 2HR • Q5 • 3 marks

Shane invests 7200 dollars for 3 years at 2.5% per year compound interest. How much will he have at the end of the 3 years? (nearest dollar)

STEP 1 $m = 1.025$ ◀ multiplier

STEP 2 1.025^3 ◀ 3 years → power 3

STEP 3 $7200 \times 1.025^3 = 7753.6 \dots$ ◀ multiply

ANSWER \$7754

✓ year by year: 7380, 7564.50, 7753.61 ✓

WATCH OUT

✗ $7200 \times 0.025 \times 3 + 7200 = 7740$

✓ $7200 \times 1.025^3 = 7754$

That is **simple** interest. Compound means interest on the interest.

EXAM TIP

11 of the 29 past-paper parts. Write 7200×1.025^3 before the answer — it is the method mark.

Check yourself

calculator first • answers on page 15

1.1 Jun 2024 P2H

4500 koruna for 4 years at 2.4% compound interest. Final amount (nearest koruna)?

Answer:

1.2 Jun 2025 P2H

A bracelet costs 4000 Swiss francs and increases by 7% each year. Value after 3 years?

Answer:

1.3 Oct/Nov 2020 P1HR

A clock bought for £1200 increases by 8% per year. Value after 3 years (nearest £)?

Answer:

V2

VARIANT 2

Depreciation

The value falls by the same percentage each year: multiplier below 1.

SEEN IN PAST PAPERS

6 / 29

PRACTICE QUESTIONS: 1



Your notes

Came up in: Jan 2020 P2HR Q8 • Jun 2022 P1HR Q8 • Jun 2022 P1H Q6(b) • May 2023 P1HR Q9 • Nov 2023 P2H Q7 • Nov 2025 P2H Q5

? What is it?**In plain words**

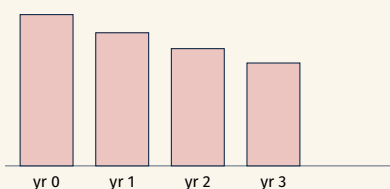
Depreciates by 12% → each year it keeps **88%** → $\times 0.88$.

Same formula: $P \times 0.88^n$. "Total percentage fall" over 2 years: $1 - 0.88^2$.

IT LOOKS LIKE

car depreciates by 19% each year

boat loses 8%



$\times 0.88$ each year: each fall is smaller than the last

How do I do it?**1 Multiplier**

-12% → 0.88.

2 Power

Number of years.

3 Multiply

$P \times 0.88^3$.

4 Total % fall

$(1 - m^n) \times 100$ if asked for the overall percentage.

KEEPS 88% → $\times 0.88$ Not $\times 0.12$.

V2 Depreciation

VARIANT 2 · SEE IT DONE, THEN TRY

 Your notes
WHEN YOU SEE → YOU DO

"depreciates in value by 19% each year" → $\times 0.81^n$

"total percentage by which the value had depreciated" → $(1 - 0.85^2) \times 100$

"nearest taka / nearest \$" → **Round only at the end**

WORKED EXAMPLE **PAST PAPER**

May 2023 · Paper 1HR · Q9 · 3 marks

Kazi buys a car for 700 000 taka. It depreciates by 12% each year. Work out the value at the end of 3 years (nearest taka).

STEP 1 $m = 1 - 0.12 = 0.88$ ◀ multiplier

STEP 2 0.88^3 ◀ 3 years

STEP 3 $700\,000 \times 0.88^3 = 477\,030.4$ ◀ multiply

ANSWER 477 030 taka

✓ it falls each year ✓

WATCH OUT✗ $700\,000 \times 0.12^3$ ✓ $700\,000 \times 0.88^3$

0.12 is what is lost; the car keeps 0.88.

EXAM TIP

Overall percentage after 2 years at 15%:

 $1 - 0.85^2 = 0.2775$, i.e. 27.75% — not 30%.**Check yourself**

calculator first · answers on page 15

2.1 Jan 2020 P2HR

A \$20 000 car depreciates by 19% each year. Value after 3 years (nearest \$)?

Answer:

2.2 Nov 2023 P2H

A \$26 800 boat depreciates by 8% each year. Value after 3 years?

Answer:

2.3 Jun 2022 P1H

A boat depreciates by 15% each year. Total percentage depreciation after 2 years?

Answer:

V3**VARIANT 3**

Compare Options

Work out every option fully (same units, same years), then compare.

SEEN IN PAST PAPERS

2 / 29

PRACTICE QUESTIONS: 6



Your notes

Came up in: Nov 2021 P2H Q7 · Jun 2025 P1HR Q11

? What is it?

In plain words

Two banks or offers. Compute the **interest** (or final value) for **each**, then answer “which?” or “how much more?”.

Simple interest: same amount each year (1150×3). **Compound**: $P \times m^n - P$.

IT LOOKS LIKE

Cyclone vs Tornado bank

ratio of money : interest

CYCLONE

4.5% compound
 $25\,000 \times 1.045^3$
 interest **3529.15**

TORNADO

1150 each year
 1150×3
 interest **3450**

\$79 MORE

How do I do it?

1 Option A

Final value or interest, fully.

2 Option B

Same quantity, same number of years.

3 Compare

Subtract (how much more) or pick the bigger.

4 Answer

With units and rounding asked.

SAME QUANTITY FOR BOTH Interest vs interest, total vs total.

V3 Compare Options

VARIANT 3 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Receive 1150 dollars interest each year”

→ **Simple:** 1150×3

“Ali will have received more interest ... how much more?”

→ **Interest A – interest B**

WORKED EXAMPLE **PAST PAPER**

November 2021 · Paper 2H · Q7 · 4 marks

Cyclone Bank: 25 000 dollars at 4.5% compound interest for 3 years.
Tornado Bank: 1150 dollars interest each year for 3 years. How much more interest does Ali (Cyclone) get? (nearest dollar)

STEP 1 $25\,000 \times 1.045^3 - 25\,000 = 3529.15$ ◀ Cyclone interest

STEP 2 $1150 \times 3 = 3450$ ◀ Tornado interest

STEP 3 $3529.15 - 3450 = 79.15$ ◀ compare

ANSWER \$79 more

✓ interest compared with interest ✓

WATCH OUT

✗ $28\,529.15 - 3450$

✓ $3529.15 - 3450$

Compare interest with interest — not a total with an interest.

EXAM TIP

Write a label on each line (“Cyclone interest”) — it stops mix-ups and earns method marks.

Check yourself

calculator first · answers on page 15

3.1 Jun 2025 P1HR

Bank A: \$2500, money : interest after 2 years = 20 : 3. Bank B: \$3000 at 4% compound for 2 years. How much more from A?

Answer:

3.2 Dr Eslam

£1000 earns 5% in year 1 and 2% in year 2 (compound). Total after 2 years?

Answer:

3.3 Dr Eslam

£2000 for 3 years: 5% simple or 5% compound. Which gives more interest, and by how much?

Answer:

V4

VARIANT 4

Find the Start Value

You know the value after the change:
divide by the multiplier (to the power).

SEEN IN PAST PAPERS

3 / 29

PRACTICE QUESTIONS: 3


Your notes

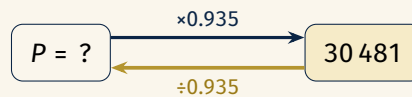
Came up in: Jan 2021 P1H Q7(b) • May 2021 P2H Q11(a) • Jun 2022 P1H Q6(a)

? What is it?

In plain words

Reverse percentages again: $P \times m^n =$
value, so $P = \text{value} \div m^n$.

After 1 year of -6.5%: 30 481 is 93.5%
of $P \rightarrow P = 30\,481 \div 0.935$.



IT LOOKS LIKE

the amount he paid

value in 2020

How do I do it?

1 Multiplier

-6.5% $\rightarrow$ 0.935; +24% $\rightarrow$ 1.24.

2 Years

n years $\rightarrow m^n$.

3 Divide

$P = \text{value} \div m^n$.

4 Check forwards

$P \times m^n$ gives the value back.

BACKWARDS = DIVIDE As in Percentages V4.

V4 Find the Start Value

VARIANT 4 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“At the end of the first year the value was 30 481. Work out P ”

$$\rightarrow P = 30\,481 \div 0.935$$

“sold for 180 000, which is 24% more than he paid”

$$\rightarrow 180\,000 \div 1.24$$

“Has the band changed?”

$\rightarrow$ **Find last year's value, compare bands**

WORKED EXAMPLE **PAST PAPER**

January 2021 · Paper 1H · Q7(b) · 3 marks

Wang invested P yuan. The value decreased by 6.5% each year. At the end of the first year it was 30 481 yuan. Work out P .

STEP 1 $m = 1 - 0.065 = 0.935$ $\leftarrow$ multiplier

STEP 2 $P \times 0.935 = 30\,481$ $\leftarrow$ one year

STEP 3 $P = 30\,481 \div 0.935 = 32\,600$ $\leftarrow$ divide

ANSWER $P = 32\,600$

$$\checkmark 32\,600 \times 0.935 = 30\,481 \checkmark$$

WATCH OUT

$$\times 30\,481 \times 1.065$$

$$\checkmark 30\,481 \div 0.935$$

Adding 6.5% back uses the wrong amount. Divide by the multiplier.

EXAM TIP

Always check forwards — if it doesn't give the value in the question, you went the wrong way.

Check yourself

calculator first · answers on page 15

4.1

May 2021 P2H

A house sold for 180 000 yuan, 24% more than was paid. What was paid (3 s.f.)?

Answer:

4.2

Jun 2022 P1H

An apartment is 634 400 euros, 4% up on 2020. Band B is 600 000–700 000. Has its band changed?

Answer:

4.3

Dr Eslam

After 2 years at 3% compound interest, an account holds £5304.50. How much was invested?

Answer:

V5

VARIANT 5

Find the Rate

The percentage is unknown: build the equation $P \times m \dots = \text{value}$, solve for the multiplier.

SEEN IN PAST PAPERS

7 / 29

PRACTICE QUESTIONS: 8



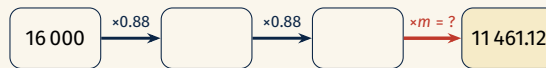
Your notes

Came up in: Jan 2020 P1H Q11 • Oct/Nov 2020 P1H Q13 • Jan 2021 P2HR Q14 • Jan 2022 P1H Q11 • Jun 2023 P2HR Q13 • Jun 2025 P1H Q13 • Jun 2025 P2HR Q12

? What is it?**In plain words**

Write the **chain of multipliers** with the unknown one as m .

Divide out the known ones to get m (or m^n). Take a **root** if there is a power. Then $x = (m - 1) \times 100$ — or $(1 - m) \times 100$ for a fall.



$$m = \frac{11\,461.12}{16\,000 \times 0.88^2} = 0.925$$

IT LOOKS LIKE

$$16\,000 \times 0.88^2 \times \left(1 - \frac{x}{100}\right) = 11\,461.12$$

**How do I do it?****1 Chain**

Start $\times$ every multiplier = final; unknown one is m .

2 Isolate m

Divide the final by everything known.

3 Root if needed

$$m^6 = 1.1097 \Rightarrow m = \sqrt[6]{1.1097}.$$

4 Rate

$$x = (m - 1) \times 100 \text{ (growth) or } (1 - m) \times 100 \text{ (fall).}$$

SOLVE FOR THE MULTIPLIER

Then turn it into a percentage.

V5 Find the Rate

VARIANT 5 · SEE IT DONE, THEN TRY

 Your notes
WHEN YOU SEE → YOU DO

"x% for the second and third year ... work out x" → **Chain with m^2 , isolate, square-root**

"total interest he receives is 6151.25" → **Final = start + interest first**

"Give your answer to 2 decimal places" → **Keep the full root until the end**

WORKED EXAMPLE **PAST PAPER**

June 2025 · Paper 2HR · Q12 · 3 marks

A \$16 000 car depreciates by 12% each year for 2 years, then by x% in the third year. Its value after 3 years is \$11 461.12. Work out x.

STEP 1 $16\,000 \times 0.88^2 \times m = 11\,461.12$ ◀ the chain

STEP 2 $m = \frac{11\,461.12}{12\,390.40} = 0.925$ ◀ isolate the multiplier

STEP 4 $x = (1 - 0.925) \times 100$ ◀ a fall

ANSWER $x = 7.5$

$$\checkmark 16\,000 \times 0.88^2 \times 0.925 = 11\,461.12 \checkmark$$

WATCH OUT

✗ $m = 0.925 \Rightarrow x = 92.5$

✓ $x = (1 - 0.925) \times 100 = 7.5$

m is what the car **keeps**. The rate is what it loses.

EXAM TIP

Use **SHIFT** **CALC** on the whole equation to check your x in seconds.

Check yourself

calculator first · answers on page 15

5.1 Jan 2020 P1H

\$6000 at 1.5% compound for 2 years, then a new rate for year 3; final \$6311.16. Year-3 rate (1 d.p.)?

Answer:

5.2 Jun 2025 P1H

\$750 000 falls 4% then 6.5%, then rises x%: now \$698 445. Find x.

Answer:

5.3 Oct/Nov 2020 P1H

\$8000 at x% compound: \$8877.62 after 6 years. Find x (2 d.p.).

Answer:

CHECK YOURSELF

Answers



Your notes

Every answer below was checked with sympy (check_answers.py).

1.1 4948 koruna**1.2** 4900 Swiss francs**1.3** £1512**2.1** \$10 629**2.2** \$20 869**2.3** 27.75%**3.1** \$130.20 more**3.2** £1071**3.3** compound, by
£15.25**4.1** 145 000 yuan**4.2** No: 610 000 and
634 400 are both band B**4.3** £5000**5.1** 2.1%**5.2** $x = 3.75$ **5.3** $x = 1.75$

THE WHOLE TOPIC ON ONE PAGE

Compound Interest cheat sheet



Your notes
V1 Growth 11/29

$$7200 \times 1.025^3 = 7754$$

$$P \times m^n, m = 1 + \frac{r}{100}.$$

V2 Depreciation 6/29

$$700\,000 \times 0.88^3$$

$$m = 1 - \frac{r}{100}.$$

V3 Compare 2/29

$$3529.15 - 3450 = 79.15$$

Same quantity for both options.

V4 Start value 3/29

$$30\,481 \div 0.935$$

Backwards: divide by m^n .**V5 Find the rate 7/29**

$$m = \frac{11\,461.12}{16\,000 \times 0.88^2}$$

Isolate m , root, then the %.**TOP 3 LOST MARKS**

1. simple interest instead of compound ×
2. $\times 0.12$ for a 12% fall ×
3. giving m instead of the rate ×

BOTTOM LINE One multiplier per year $\rightarrow P \times m^n$ • backwards: divide •
unknown rate: solve for m .